Bach Phan-Tat

phantatbach.github.io github.com/phantatbach phantatbach(AT)gmail.com

Data scientist with a background in finance and linguistics, currently a PhD candidate in computational linguistics. Skilled in quantitative analysis and machine learning, seeking opportunities to grow in AI and data science and apply data-driven methods to real-world problems.

WORK EXPERIENCE

The Digital Humanities Institute, University of Sheffield **Sep 2026 – Nov 2026**
Visiting Research Scholar *Sheffield, England*

Murf AI (speech technology company) **Nov 2024 – Present**
Remote Research Consultant *Utah, USA*
Project: Multilingual Text-to-Speech system

- Developed phonetic inventories and pronunciation lexicons (word–IPA mappings) for Vietnamese, Turkish, and multiple English varieties (US, UK, Scottish, and Australian), applying corpus-linguistic, phonological, and statistical methods to support multilingual TTS system development.
- Designed evaluation pipelines and test strategies for sentence segmentation models.
- Implemented Python scripts for automated text processing, lexical analysis, and language data preparation.
- Performed linguistic error analysis, documented findings, and recommended improvements for data and model quality.
- Reviewed relevant literature and industry practices to support solution design for multilingual speech technology tasks.
- Worked with cross-functional teams on annotation, transcription, and data preparation for speech and language workflows.

Vbee (speech technology and conversational AI company) **Feb 2023 – Aug 2024**
Junior Research Engineer *Hanoi, Vietnam*
Project: HUST Chatbot (a conversational AI for Hanoi University of Science and Technology)

- Developed and evaluated machine learning and BERT-based language models for classification tasks, including gender classification from names, chat domain classification, and toxicity detection, achieving accuracies above 85%.
- Designed statistical tests for data sampling and human evaluation.
- Developed a chatbot evaluation pipeline using commercial LLMs to assess response quality and system performance.
- Built and maintained language data workflows involving crawling, extraction, LLM-based data generation, pre-processing, annotation, and quality review for text and audio datasets.
- Contributed to the development of linguistic resources, prompts, and rule-based components for speech and language tasks, including speech-to-text, text-to-speech, and part-of-speech parsing.

what3words (location-mapping technology company) **Jul 2023 – Aug 2023**
Remote Research Consultant *London, England*
Project: Vietnamese Automatic Speech Recognition system (in collaboration with VinFast)

- Conducted linguistic error analysis of failed ASR outputs, identifying systematic patterns related to phonetic confusion and semantic ambiguity.
- Proposed data- and model-level improvements by combining linguistic analysis with machine learning insights.

ACADEMIC & PROFESSIONAL SERVICE

Supervising

- Advised a Master's student at Lancaster University on a Linguistics dissertation, which received a Distinction (76%). 2024.

Mentoring

- Mentored an applicant who was later accepted to the YSEALI Fellowship. 2024.
- Mentored an applicant who was later accepted to the PhD programme in Linguistics at Lancaster University. 2024.

Reviewing

- Reviewer for Cognitive Linguistic Studies. 2026.
- Reviewer for Linguists' Day, Linguistic Society of Belgium (LSB). 2025.

Organising

- Co-organised the PhD Day for the Department of Linguistics at KU Leuven. 2026.

PUBLICATIONS

Book chapters (peer-reviewed)

- **Bach Phan-Tat***, Sofía Aguilar-Valdez*, Stefania Degaetano-Ortlieb, Dirk Geeraerts, & Dirk Speelman. 2025. Discursive parallels of the chemical revolution: topic modelling and distributional analysis. In *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion* (open access, transcript). (Book chapter / edited volume). *Equal contribution.

Conference and workshop proceedings (peer-reviewed)

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. ReFRAME or Remain: Unsupervised Lexical Semantic Change Detection with Frame Semantics. *Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (*SEM 2026), ACL 2026.*
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Transparent Semantic Change Detection with Dependency-Based Profiles. *Proceedings of the 6th International Workshop on Computational Approaches to Historical Language Change (LChange'26), EACL 2026.*
- Khai Le-Duc, Tuyen Tran, **Bach Phan-Tat**, et al. 2025. MultiMed-ST: Large-scale Many-to-many Multilingual Medical Speech Translation. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025).*
- Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, **Bach Phan-Tat**, Minh-Huong Ngo, Thanh Nguyen-Tang, Truong-Son Hy. 2025. MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025).* Top 15% of accepted papers.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, Le Duy, Long Vo-Dang, Truong-Son Hy. 2025. Sentiment Reasoning for Healthcare. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025).* Oral presentation.

Pre-prints and under-review articles

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. Evaluating the Evaluator: Problems with SemEval-2020 Task 1 for Lexical Semantic Change

Detection (Preprint - Under review)
- Catherine Wong, **Bach Phan-Tat**, Susan Fitzmaurice. 2026. Methods, Data, and Conceptual Change: Reflections from Two Quantitative Diachronic Case Studies (Preprint - Under review)
- Duc Cao-Dinh, Khai Le-Duc, Anh Dao, **Bach Phan-Tat**, Chris Ngo, Duy M. H. Nguyen, Nguyen X. Khanh, Thanh Nguyen-Tang. 2025. Audio-3DVG: Unified Audio - Point Cloud Fusion for 3D Visual Grounding. (Preprint).

PRESENTATIONS

Conference presentations
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Underlying Dimensions of Semantic Change. Digital Humanities Congress 2026.
- **Bach Phan-Tat**, Kris Heylen, and Stefano De Pascale. 2026. Lexical Semantic Change Detection? A Good Parser Might Be Enough! 1st LOT Conference.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. SynFlow: Continuous Semantic Change Analysis via Dependency Co-occurrences. ICAME 47.
- **Bach Phan-Tat**, Dirk Geeraerts, and Dirk Speelman. 2026. Conceptual Change during the Chemical Revolution: Air, Acid, and Water in the Royal Society Corpus. ICAME 47.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. From Parsers to Prompts: Combining AI and Linguistics for Interpretable Language Evolution. AI in Language Evolution Workshop, Evolang 2026.
- **Bach Phan-Tat**. 2026. Slot-Filler Distributional Changes in Times of Scientific Debate: A Case Study of Air. Quantitative Diachronic Linguistics and Cultural Analytics.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. 2024. Sentiment Reasoning for Healthcare. Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond Workshop, NeurIPS 2024.

Invited talks & seminars
- **Bach Phan-Tat**. 2026. Underlying Dimensions of Semantic Change. NLP & text mining seminar organised by Leuven.AI.
- **Bach Phan-Tat**. 2026. SynFlow: Continuous Semantics Change Analysis via Dependency Co-occurrences. Data in Historical Linguistics seminar series.

EDUCATION

KU Leuven **Sep 2024 – Present**
PhD, Computational Linguistics *Leuven, Belgium*
Thesis: **Underlying Dimensions in Conceptual Change**
Selected Achievements
- Developed a general, lightweight, and interpretable framework for lexical semantic change detection and analysis, using off-the-shelf parsers or LLMs to extract structured linguistic patterns and Jensen-Shannon Divergence to quantify diachronic distributional shifts.
- Developed SynFlow, a Python package for detecting and analysing semantic change through syntactic co-occurrence patterns and Jensen–Shannon Divergence.
  - Implemented efficient syntactic pattern extraction and structuring using BFS/DFS, n-ary tree search, and rule-based filtering.

  - Optimised I/O-bound batch processing tasks, including multi-file search and DataFrame construction from large collections of corpus files, using multithreading.
- Built automated linguistic annotation pipelines using local LLMs for corpus-based analysis.
- Cleaned, parsed, and lemmatised large diachronic and contemporary text corpora for downstream semantic analysis.
- Optimised computational workflows through parallel, batch, and distributed processing, reducing processing time by 80%.
- Maintained and extended NephoSem, a lexical semantics toolkit developed by the QLVL research unit.
  - Adapted the pipeline to support multiple word embedding types for downstream analysis.

Thesis supervisors: Prof. Dr. Dirk Speelman, Prof. Dr. Dirk Geeraerts, Dr. Kris Heylen

WorldQuant University **Oct 2025 – Present**
MSc, Financial Engineering *Remote*
Finished modules: Financial Market, Financial Data

University of Stirling **Sep 2021 – Sep 2022**
MSc, English Language and Linguistics *Stirling, Scotland*

Thesis: **The structure of *thời gian*: A semantic analysis of the Vietnamese lexeme *thời gian* based on Principled Polysemy**
Selected Achievements

- Randomly sampled, crawled, cleaned, and compiled a specialised corpus of 500 Vietnamese news articles using Python and #LancsBox.
- Identified seven senses of the Vietnamese lexeme *thời gian* ('time') and traced their semantic development using the Principled Polysemy framework.

Thesis supervisor: Dr. Andrew Smith
Thesis grade: 80% (Distinction; class rank: 1st)
Average grade: 78% (Distinction; class rank: 1st)

Academy of Finance **Aug 2016 – Aug 2020**
BA, Corporate Finance *Hanoi, Vietnam*
Relevant modules: Linear Algebra, Calculus, Probability and Statistics, Econometrics

TECHNICAL SKILLS

Programming & Computing

- Programming languages (ordered by proficiency): Python (NumPy, Pandas, Matplotlib, scikit-learn, spaCy, PyTorch, Hugging Face), R, Bash, LaTeX, MATLAB, Java
- Parallel, batch, and GPU-enabled computing workflows
- Databases: MongoDB, SQL
- Git/GitHub

Data Science

- Exploratory data analysis, data cleaning and (interactive) data visualisation
- Machine learning and deep learning for NLP, time-series, structured data and computer vision
- Quantitative research design, experimental design, and A/B testing
- LLM-based annotation and evaluation workflows
- Mathematics for data science: linear algebra, multivariate calculus, statistics

Linguistics & Corpus Methods

- Corpus tools: CQPweb, #LancsBox, Sketch Engine, AntConc
- Linguistic analysis: phonetics, morphology, syntax, semantics
- Linguistic annotation, transcription, and corpus preparation

AWARDS AND HONOURS

- 2026: 1st place, Databricks Challenge (USD 1,000); 4th Hack-Nation Hackathon.
- 2024: Marie Skłodowska-Curie Actions Doctoral Fellowship (3 years), Horizon Europe.
- 2022: Research-Based Learning Prize for the best Master's dissertation in Literature and Languages, University of Stirling (GBP 225); first student in the history of the Linguistics Department to receive the award.
- 2021: Postgraduate Scholarship, University of Stirling (GBP 4,000).

SELECTED ADDITIONAL TRAINING

Data Science
- Inferential Statistics (The University of Amsterdam)
- Applied Data Science Lab (WorldQuant University)
- Deep Learning (Neuromatch Academy)
- Applied AI Lab (WorldQuant University)

Linguistics
- Corpus Linguistics: Method, Analysis, Interpretation (Lancaster University)
- Corpus Linguistics Summer School (Birmingham University)

Financal Engineering
- Foundations of Financial Engineering (WorldQuant University)

Research Methodology
- Quantitative Method (The University of Amsterdam)
- Qualitative Research Methods (The University of Amsterdam)
- Open Science 101 (Neuromatch Academy x NASA)

LANGUAGES

- Vietnamese: Native
- English: IELTS 8.5 – C2
- Dutch: A2